

A.5 Key Discrete Distributions

Loa Geometric Series

- many discrete distributions can have a non-negative integer. deriving mean & var. of such requires infinite series expansion.

- If $|a| < 1$, then

$$S = \frac{a}{1-a}$$

- In general, sum of geometric series is,
 $S = \frac{\text{first term}}{1 - \text{ratio}}$

$$\sum_{n=0}^{\infty} ar^n = \frac{ar^0}{1-a}$$

- for a partial sum (the final result holds true for any n)

$$\sum_{n=0}^m ar^n = \frac{\text{first term} - \text{first missing term}}{1-a}$$

eg. $\sum_{n=3}^{\infty} 5 \cdot \frac{e^{n+2}}{2^{3n} 3^{-n}}$

first term = 3

first missing = 18

$$r = \frac{e}{2^3 3^{-1}} = \frac{3e}{8}$$

$$= \frac{5e^5}{2^9 3^{-3}} - \frac{5e^{20}}{2^{318} 3^{-18}}$$

$$1 - \frac{3e}{8}$$

$$S = \sum_{n=0}^{\infty} n \cdot a r^n, \quad |a| < 1$$

$$S = \frac{ar}{(1-a)^2}$$

→ this can be used to find the mean of a geometric random variable.

- for $S = \sum_{n=0}^{\infty} n^2 a r^n \Rightarrow S = \frac{ar(1+a)}{(1-a)^3}$

- for $S = \sum_{n=0}^{\infty} n^2 a r^n$, by calculus, also, you'll achieve the same formula.

Lo b Taylor Series

$$1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} = e^x$$

$$\text{for } e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

Eq. $\sum_{n=0}^{\infty} \frac{5^n e^{tn}}{n!} \cdot e^{-5} = e^{-5} \cdot \sum_{n=0}^{\infty} \frac{(5e^t)^n}{n!}$

$= e^{-5} \cdot \sum_{n=0}^{\infty} \frac{u^n}{n!} = e^{-5} \cdot e^u = e^{-5} \cdot e^{5e^t}$

$= e^{5e^t - 5}$

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eg. $\sum_{n=2}^{\infty} \frac{2^n}{n!} = ?$, but $\sum_{n=0}^{\infty} \frac{2^n}{n!} = e^2$

$$e^2 - \left(\frac{1}{0!} + \frac{2}{1!} \right) = e^2 - 3$$

L1

Geometric Distribution

- suppose I roll a die until I get 6. Let N be the total no. of rolls. What is the distribution of N ?

$$P(N > 0) = 1$$

$$P(N=1) = \frac{1}{6}$$

$$P(N > 1) = \frac{5}{6}$$

$$P(N=2) = \left(\frac{5}{6}\right) \cdot \frac{1}{6}$$

$$P(N > 2) = \left(\frac{5}{6}\right)^2$$

$$P(N=3) = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}$$

$$P(N > k) = \left(\frac{5}{6}\right)^k$$

$$P(N=n) = \left(\frac{5}{6}\right)^{n-1} \cdot \frac{1}{6}$$

this is a geometric RV.

$$\begin{aligned} - E[N] &= \sum_{n=1}^{\infty} n \cdot P(N=n) = \sum_{k=0}^{\infty} P(N > k) \\ &= \sum_{n=1}^{\infty} n \times \frac{1}{6} \times \left(\frac{5}{6}\right)^{n-1} = \sum_{k=0}^{\infty} \left(\frac{5}{6}\right)^k \end{aligned}$$

- Geometric starting at 1 $\{1, 2, \dots\}$

$$P(X=n) = p(1-p)^{n-1}$$

$$E[X] = \frac{1}{p}$$

$$\text{Var}(X) = \frac{1-p}{p^2}$$

↳ here X is the no. of trials up to & including the first success.

- Geometric starting at 0 $\{0, 1, 2, \dots\}$

$$P(Y=n) = p(1-p)^n$$

$$E[Y] = \frac{1-p}{p}$$

$$\text{Var}(Y) = \frac{1-p}{p^2}$$

↳ no. of failures before the first success.

Ex. Let $N(0)$ randomly chosen insured makes up to doctor. N is geometric with mean 3. $P(N \geq 2)$

$$3 = \frac{1-p}{p} \Rightarrow 3p = 1-p \Rightarrow 4p = 1 \Rightarrow p = \frac{1}{4}$$

$$P(N \geq 2) = 1 - P(N < 2)$$

$$= 1 - P(N=0) - P(N=1)$$

$$P(N=0) = p(1-p)^0 = \frac{1}{4} \left(\frac{3}{4} \right)^0 = \frac{1}{4} \times \frac{4}{4} = \frac{1}{4}$$

$$P(N=1) = p(1-p)^1 = \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$$

$$1 - \frac{1}{4} - \frac{3}{16} = \frac{4}{16} - \frac{3}{16} = \frac{1}{16}$$

L2 Memoryless Property

- Let N be a game length
we start from 3 rounds, so $N-3$ rounds to watch.

$$N > 3.$$

$$P(N-3 = k \mid N > 3) = P(N = k)$$

so $(N-3 \mid N > 3)$ and original N , both have the same distribution.

$$E[N-3 \mid N > 3] = E[N] = \text{same ans.}$$

- So, if N is a geometric with parameter p , then $(N-k \mid N > k)$ is a geometric starting at 1 with same p .
 - $(N-k \mid N > k)$ is at least 1, hence starts at 1.
 - Holds true for 0 & 1.
 - Geometric is only discrete distribution with this property.

Ex. $X \rightarrow P(X=k) = 0.2(0.8)^k$ for $0, 1, \dots$
 $E[X \mid X > 6] = ?$, $\text{Var}(X \mid X > 6) = ?$

$$P(X=k) = 0.2(0.8)^k \quad 0, 1, \dots$$
$$\approx p(1-p)^n$$

$$p = 0.2$$

$$E[X] = \frac{1-p}{p} = \frac{0.8}{0.2} = 4, \quad \text{Var}(X) = \frac{1-p}{p^2} = \frac{0.8}{0.04} = 20$$

$$E[X-6+6 \mid X > 6] = E[X-6 \mid X > 6] + 6 = 20$$
$$= 4+6 = 10+1 = 11 \quad (\text{starts at 1 and not 0})$$

L3 Negative Binomial Distribution

- Roll a die until the 3rd time you get a 6. N is no. of failures. Distribution of N ?

$N = n$, then $n+3$ was 6.

$$P(N=n) = \binom{n+(3-1)}{n} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^n$$

- So, N is a negative binomial distribution with parameters r & p if it is the sum of r independent geometric variables starting at 0. It is the no. of failures before the r^{th} success.

$$P(N=n) = \binom{n+(r-1)}{n} \times p^r \times (1-p)^n$$

$$E[N] = r \frac{(1-p)}{p}$$

$$\text{Var}(N) = r \frac{(1-p)}{p^2}$$

eg. $P(A \geq 1) = \frac{3}{5}$, find the probab. that June will be 4th month in which at least 1 accident occurs.

$$p = \frac{2}{5}, \quad n = 4^{\text{th}} \text{ in } 6^{\text{th}} \text{ try}$$

$$n = 6 - 4 = 2$$

$$\binom{2 + (4-1)}{4-1} \times \left(\frac{2}{5}\right)^4 \times \left(\frac{2}{5}\right)^2$$

Eg. N is the sum of n indep. geom. $(0, 1, 2, \dots)$.
 $E[N] = 12$, $\text{Var}(N) = 60$. $P(N \leq 2)$

$$E[N] = \frac{n(1-p)}{p} = 12 \Rightarrow \frac{12}{p} = 12 \Rightarrow$$

$$\text{Var}(N) = \frac{n(1-p)}{p} \times \frac{1}{p} = 60 \Rightarrow \frac{12}{p} = 60 \Rightarrow$$

$$p = \frac{1}{5}, \quad n = 3$$

$$\begin{aligned} P(N \leq 2) &= P(N=0) + P(N=1) + P(N=2) \\ &= p^3 + {}^3C_2 (1-p)p^3 + {}^4C_2 (1-p)^2 p^3 \\ &= 0.05792 \end{aligned}$$

L4 Poisson Distribution

- X is a poisson (λ) RV if,
 $P(X=n) = e^{-\lambda} \frac{\lambda^n}{n!}, \quad n=0, 1, 2, \dots$

↳ it arises by counting the no. of occurrences of unusual events if the no. of occurrences in disjoint time intervals are independent.

$$E[N] = \lambda$$

$$\text{Var}(N) = \lambda$$

$$P(N=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

→ the probability mass funcⁿ.

eg. $P(N=2) = 3 \cdot P(N=4)$, $\text{Var}(N) = ?$

$$\frac{e^{-\lambda} \cdot \lambda^2}{2!} = 3 \times \frac{e^{-\lambda} \cdot \lambda^4}{4!}$$

$$\frac{4!}{6} = \lambda^2 \Rightarrow \lambda^2 = 4$$

$$\lambda = 2 = \text{Var}(N)$$

eg. $N \sim \text{Poisson}(\lambda)$, $E[N^2] = 12$

$$P(N \geq 2) = ?$$

$$\lambda = E[N^2] - \lambda^2$$

$$\lambda^2 + \lambda - 12 = 0 \Rightarrow (\lambda + 4)(\lambda - 3) = 0$$

$$\lambda = 3, -4$$

$$P(N \geq 2) = 1 - P(N=0) - P(N=1)$$

$$= 1 - e^{-3} - 3e^{-3}$$

$$= 0.8$$

LS: Sum of Poisson Variables

eg. $X \sim \text{Poisson}(1.7)$, $Y \sim \text{Poisson}(1.3)$

$$E[X+Y] = E[X] + E[Y]$$

$$= 1.7 + 1.3 = 3$$

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$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y), \text{ by independence}$$

$$= 1.7 + 1.3 = 3$$

$$P(X+Y=2), \quad (X+Y) \sim \text{Poisson}(3)$$

$$P(Z=2) = \frac{e^{-\lambda} \lambda^2}{2!} = e^{-3} \frac{3^2}{2} = \frac{9}{2} e^{-3}$$

- So, if $N \sim \text{Poisson}(\lambda)$, $M \sim \text{Poisson}(\mu)$ & M & N are independent.

$$P(N+M=n) = P(\text{Poisson}(\lambda+\mu) = n)$$

- If $N_1, \dots, N_k$ are independent Poissons, then their sum is Poisson.

Eg. $\text{Weekday} \sim \text{Poisson}(0.5)$
 $\text{Weekend} \sim \text{Poisson}(0.3)$

$$\begin{aligned} \text{No. of accident} &= 5 \times 0.5 + 2 \times 0.3 = 3.1 \\ P(N=3) &= \frac{e^{-\lambda} \lambda^n}{n!} \\ &= \frac{e^{-3.1} (3.1)^3}{3!} = 0.224 \end{aligned}$$